

Problem H. Disjoint LIS

Time Limit 2000 ms
Mem Limit 524288 kB

Description

Let the LIS of a permutation be the length of its longest increasing subsequence.

A permutation is good if it is possible to find two increasing subsequences of length LIS that do not share any common elements.

Given n , find the number of good permutations with n elements. As the answer may be large, you only need to find it modulo 998 244 353.

Input

The first line of input contains one integer n ($1 \leq n \leq 75$): the number of elements.

Output

Output one integer: the number of good permutations with n elements, modulo 998 244 353.

Sample 1

Input	Output
6	132